Chapter 1

Systems, Roles and Development Methodologies

Roles:

The Consultant:

Often an external hire. Brings a fresh, objective perspective. Lacks deep organizational knowledge. Relies on systematic methods and wer input

The Supporting Expert:

Usually an internal employee. Draws on specific technical experitise. Involved in smaller modifications. Acts as a resource for project managers.

Agent of Change:

The most comprehive and responsible role. Can be external or internal. Involved throughout the SDLC. Chage the business. Must work closely with users/management from day 1.

SDLC: Systems Development Life Cycle.

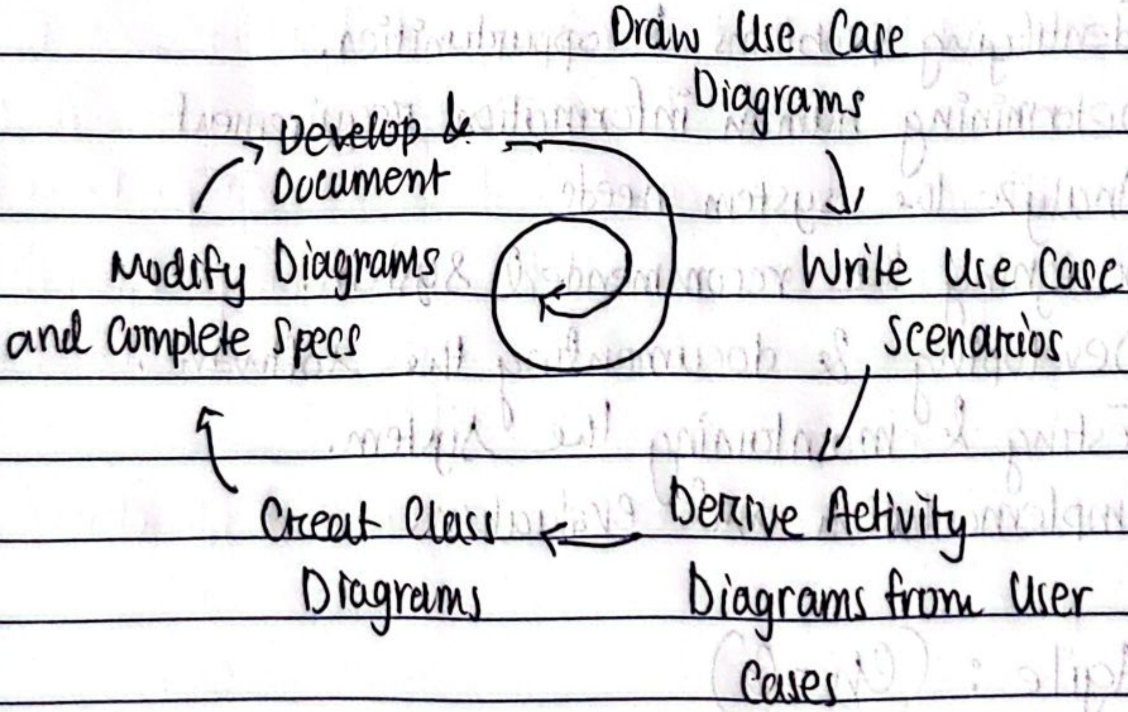
1) Identifying problems & oppurtunities.
2) Determining human information requirement
3) Analyze the system needs.
4) Designing the recommended system.
5) Developing & documenting the software.
6) Testing & maintaining the system.
7) Implementation and evaluation.

Agile: (Ch - 6)

O-O: Object - Oriented Analysis and Design

Focused on Objects grouped into Classes, defining shared attributes. Focuses on development of ~~sys~~ system that may need rapid changes and continuous adaptation. Promotes. Uses UML diagrams for modelling, Similar to SDLC due to use of rigorous UML diagrams.

O-O Workflow with UML



Creating activity / sequence diagrams often reveals need to refine / clarify use cases. ~~Draw~~ Discoveries made during class and statechart diagrams can lead to refinement of other diagrams.

Chapter 2

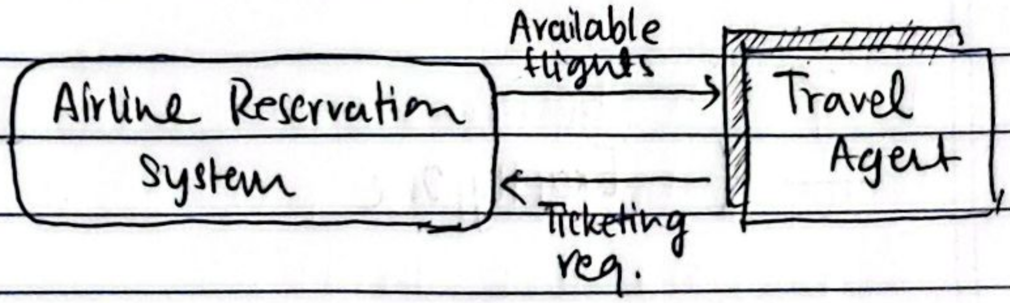
Understanding & Modelling Organizational Systems

ERP: Enterprise Resource Planning,

Large scale software system that connects the main departments of an organization into one integrated system.
Ex: Oracle Cloud, MS Dynamics.

Context Level DFD:

Shows how the system interacts with the external world. Three components
- system (rounded rect)
- external entity (shaded square)
- data flow (arrow)



Limitations:

- Shows no internal ~~details~~ details on 'how'.
- Doesnt show data storage.

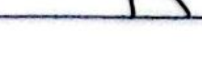
ER Models

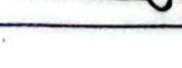
Focus on the data the system needs. Foundation of DB design.

Fundamental	Represents a basic ~~of~~ object Ex: STUDENT, DEPT, COURSE
Associative	Links to 2 or more entities, representing an event Ex: REGISTRATION (links STUDENT & COURSE)
Attributive	Describes a fundamental entity, esp for repeating groups. Ex: OFFERING (describes the sem, year, classrooms of a COURSE)

 Exactly One

 One or more

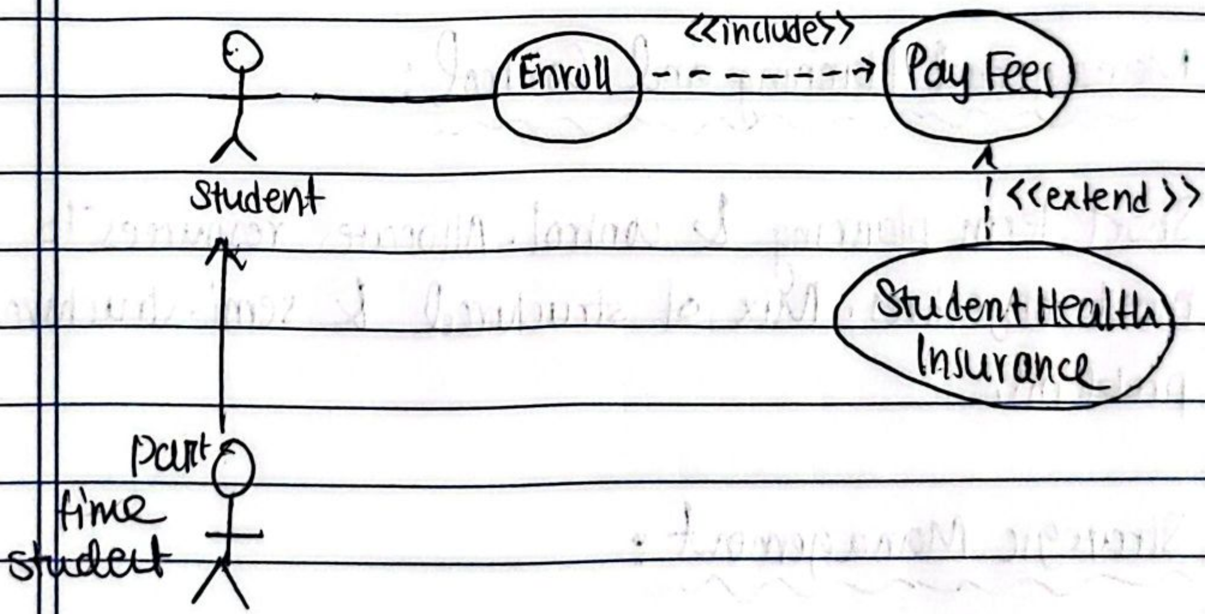
 Zero or One

 Zero, one, or more

→ Defines scope
→ Help understand the business
→ foundation of data gathering.

Use Case Modelling

Shows how actors and use cases interact with each other.



Cockburn's Altitude Metaphor

White (cloud) : Highest level (Ex: manage supply)
Kite : High level summary (Ex: manage reservation)
Blue : User goal (Ex: register student)
Indigo : functional detail (Ex: choose a class)
Black : detailed subfunction (Ex: validate login)

Three Tiers of Management

1. **Operational Control:**

directly supervises daily operations. Decides based on ~~pe~~ predetermined rules. Short-term horizone. Predictable outcomes. Repetitive decisions.

2. **Managerial Planning and Control:**

Short term planning & control. Allocates resources to meet objectives. Mix of structured & semi-structured problems.

3. **Strategic Management:**

long-term direction, looking outward at environment, setting goals. Ambiguous problems. Harder to identify. Unique, one time decisions.

Info Needs

Tier 1
Tier 2
Tier 3 ↓

Less real time detailed info needed
Less repetition. More focus on current state vs standards. More predictive info needed.

Organizational Culture

Culture influences how people work together and how decisions are made. How readily new ideas are accepted. Subcultures also present within an org. Need to identify and understand the overlapping dynamics.

Non-verbal Symbolism

verbal - Symbolism

Factors Controlled By Culture :

a) resistance to change
b) info hoarding
c) system acceptance
d) determining requirements.

3.60

FIGURE NO.:

SAD Project Management

Prob defn → repitition of theme / prob
metaphors
storytelling
Direct statement / ~~Ext~~ Emphasis

Check output / Observe Employee Behaviour / Listen to Feedback { Performace gaps, Inefficiencies, Errors

Prob defn
→ Link issues to objectives → needs follow-up interview.
→ determine the relative imp. of objectives
→ validate : experience the prob. firsthand.

~~Prob defn : Catherines Catering~~

~~Proj~~ Project Green/~~Bl~~ Red Light : Big Questions

1) Backing from Management? (endorsement from budget/resource ppl)
2) Appropriate timing? (ready for change? enough time?)
3) improves Strategic Organizational goals?
4) Resourcefull and Practical?
5) Better than alts?

Feasibility

PAGE NO.
Name of the Experiment
EXP. NO.
DATE :

Technical Feasibility :
- Can we upgrade current system?
- Does the tech exist?
- ~~Can we~~ Do we have the ~~exp~~ in-house expertise? If no, can we hire from outside?
- Are packaged solutions available?

Focus: Availability of tech & skills

Ecological Feasibility:
- ~~Do~~ Do the benefit outweigh the cost?
  - → analyst & team time + system study
  - → Training / Transition time
  - → Hardware
  - → Software / custom development cost.

Focus: Can the org see the value? is long-term gain > short term cost?

Operational Feasibility:
Will people actually use it?
Will the system operate correctly within the env?
How will it impact workflows & processes?

Watch Out For: resistance to change
lack of involvement
poor UI.

SAD Chap 2

Problems ⟶ Problem Defⁿ ⟶ List problems

Project Green/Red Light : Big Questions. ~~(7)~~ (8)

TEO Feasibility : ✓

Hardware Inventory : Assess if current staff skills match needed / existing hardware

needed / existing hardware ⟶ figure out starting point
↳ record hardware data

↳ analyzes the pros and cons of hardware
↳ benchmark ⟶ avg ~~tran~~ i/p ⟶ o/p time
↳ total ~~cap~~ throughput
↳ CPU/Network idle time
↳ mem size

↳ Buy or Rent ?

Hardware Buy / Rent

	~~Bu~~ Adv	Disad
Buy	Full control Cheaper in long Run Tax Advantage (Depraciation)	Maintenance / Operation responsibility High initial cost Risk of being stuck with wrong choice Risk of obsolescence.
Cloud.	Maintenance / upgrades by provider Lower initial cost Agility + Scalability Consistency across platforms.	Company doesnt directly control own data → potential security risk. Reliability depends on Internet or Provider. Proprietary APIs may hinder switching

Hybrid → mix

Cloud → SaaS (Gmail, MS 365) ~~(VMs, servers, load ba~~

→ PaaS (AWS, Azure → Platform to build & deploy)

→ IaaS (VMs, servers, load balancers)

Evaluating Vendor Support

- H/W → Quality / Warranty
- S/W → Budled OS/ SW ? Custom Programming ? Warranty
- Installation & Training → Commitment ? In-house training ?
- Maintenance → Routine/Preventive Procedures
  ↳ Emergency Response Time
- Disaster Recovery → 24/7 Operation ? Ransomware mitigation ? User data Migration ?

BYOD → lower initial cost + morale + flexible

↳ Design with popular platforms in mind.

↳ Security Risk.

SW Buy or Rent

- ~~cu~~ Create custom SW
- Purchase COTS
- SaaS

Cost vs Benefits

Analyze cost / benefits over time → predict future wage volume, labor cost, sales

→ Needs historical data (if available)
→ Else use judgement methods

→ regression
→ moving avg.

~~For~~ Benefits:

Tangible: Measurable in dollars.

~~Intag~~ Intangible: improved decision making
better data acc.
betterr customer service
rep ↑

Cost

Tangible: Accurately projected and quantified

Intangible: losing comp. edge
rep ↓
morale ↓

Break Even Analysis + Payback Analysis ✓

Work Breakdown Structure (WBS)

Divide and Conquer → single deliverable
→ assignable
→ accountable

Product Oriented WBS → Breaks it into components
↳ login /signup

Process " " → Breakdown in phases (SDLC phases)

Est. time req

1) Exp
2) Analogies
3) Three point est. (best + 4xavg + worst) /6
4) Function point Analysis → system's functional size & complexity
   - external i/p o/p
   - external queries
   - internal logical files
   - external interface files
5) using sw,

Gantt Chart : Doesnt show deps.

Pert Chart :
- Shows deps.
- Critical path : proj dur.
- Slack time
- Dummy Activities

Controlling Costs : Estimation

Top Down (Exp Driven)

Bottom-Up (Getting est. ~~from~~ from team)
↳ time-consuming

Parametric Modelling
↳ parameters + formulae
↳ COCOMO

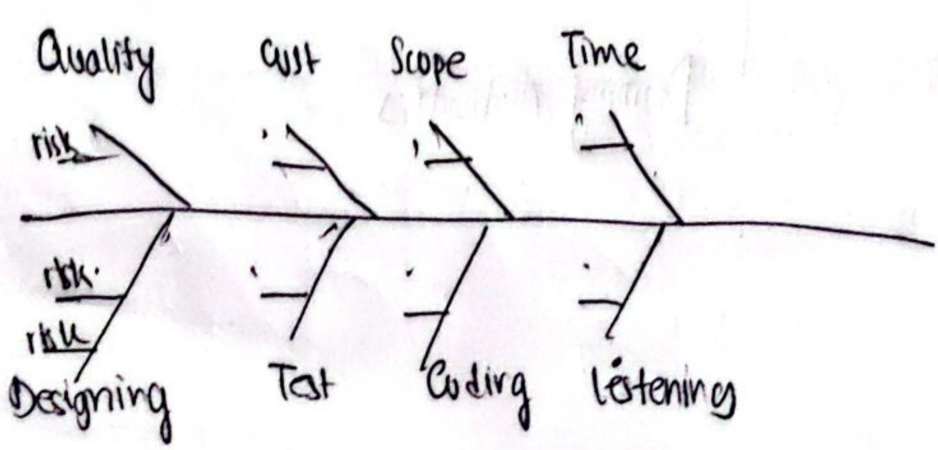
common : use combo.

Pitfalls & Budgeting

↳ Cost Estimates often fail → over optimism
↳ rushing on estimates.

Risk Management

Best Defence : ~~That~~ Thorough analysis + feasibility studies + understanding motivations + exp.



Crashing : Speeding up proj → extra cost

Expediting → done for critical path
→ pick the cheapest critical path activity
→ never expedite an activity below its crash duration.

→ Analysis in Slide

~~Ear~~ Earned Value Management (EVM)

Integrates scope, schedule & Cost into a unified framework to measure performance and predict outcomes

- Budget at completion : Total budget
- Planned Value : Spent value by proportion of $\overbrace{\text{proj}}^{\text{according to plan}}$ done
- Actual Cost : Actual ~~spend~~ spent value so far
- Earned value : BAC × Actual % completed

Formulae

BAC = $\sum_i$ Estimate Cost till end

PV = $\sum_0^i$ Estimate Cost till date of measuring

AC = $\sum_i$ Actual ~~mo~~ stage-wise cost till date of measuring

$$\%\text{ done} = \frac{\sum_i \text{stage \% completed}}{100 \times \text{no. of stages currently visited}}$$

EV = PV × % done

CV = EV − AC   CPI = EV/AC

SV = EV − PV   ~~SV~~ SPI = EV/PV

ETC = (BAC − EV) / CPI

EAC = AC + ETC